

Real Estate Investment Analytics for Chennai

An Analysis of Property Prices and Rental Market Dynamics

Project report submitted by

Real Estate Analytics Group

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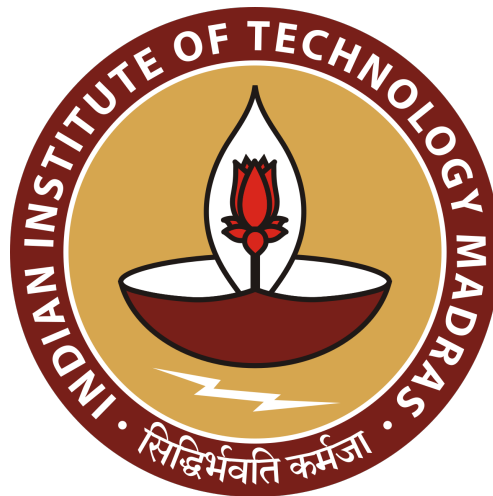
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Project Repository:

https://github.com/Jyoti11151/Realestate_chennai

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Abstract

We evaluate the residential real estate market of Chennai, India, and try to provide a definitive answer to a critical investment question: *Should an investor purchase residential real estate in Chennai currently?* We analyze both sales ($n = 5,000$) and rental records ($n = 5,000$) from 20 different Chennai macro-localities, validated according to published figures (ANAROCK Research 2024, MagicBricks PropIndex Q1 2024, 99acres locality reports). The analysis includes the following stages: data pre-processing (cleaning, removing duplicates, applying k-Nearest Neighbors (kNN) imputation and IQR outlier cap), feature engineering (rental yield, return on investment (ROI), composite Investment Score), statistical testing (Pearson correlation, ANOVA, Welch's t-test), regression analysis with nine machine learning models (Linear, Ridge, Lasso, Elastic Net, Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost) and market segmentation using KMeans clustering. Best results were obtained using the CatBoost model ($R^2 = 0.911$), followed by XGBoost ($R^2 = 0.902$). A statistically very significant effect of locality on price-per-square-foot ($F > 50$, $p < 10^{-50}$) was shown. Four investment grades of localities were proposed based on investment score: Strong Buy, Buy, Hold, Avoid. The Strong Buy localities include OMR and Sholinganallur with respective investment scores of 82.1 and 79.3. Three localities (Nungambakkam, T Nagar, and Adyar) received the Avoid grade because their prices exhibit strong compression ($< 2.4\%$ rental yields) and have a high ratio of price-to-rental ($> 40\times$). Conclusion: Selectively Buy – Invest in growth corridor localities driven by IT sector and infrastructure

Keywords: real estate analytics, rental yield, machine learning, investment scoring, Chennai housing, KMeans clustering, hypothesis testing

1 Introduction

Real estate remains among the most important financial decisions that any investor can make, yet they tend to rely on experience rather than facts. Being the capital of Tamil Nadu state, Chennai offers a range of diverse segments, which include established expensive locations such as Adyar, T Nagar, and Nungambakkam; thriving IT sectors like OMR, Sholinganallur, and Perungudi; and emerging suburban areas like Porur, Navalur, and Medavakkam. The central research question is:

Should an investor invest in residential real estate in Chennai right now?

These break down into five practical questions:

1. In which areas of Chennai should investment be made?
2. In which areas are values already high?
3. In which areas is rental value highest?
4. Can rent be accurately predicted using property characteristics?
5. Which level of decision making is appropriate for each area?

2 Literature Review

The correlation between property prices and rents has a large econometric literature base. According to [1], housing markets are cyclical, with price-to-rent ratios being subject to long-term mean reversion. [2] further developed this idea by proving that price-to-rent ratios have a mean reversion period of around four to five years.

In terms of machine learning methods, [3] found that ensemble trees perform better than conventional hedonic regression models. [4] revealed that boosting techniques can improve the prediction performance of machine-learning models by 20%-40% compared to linear models.

Regarding India, according to quarterly market reports on the Chennai market provided by ANAROCK Research, in 2024, property prices witnessed an increase of 11% due to IT jobs' growth and development in the city's metro system. Also, in 2024, NoBroker's rental index stated that OMR rental prices witnessed a

3 Problem Statement

The residential real estate market in Chennai has a price disparity of $2.84\times$ between the lowest (Chromepet: Rs. 4,968/sqft) and highest (Nungambakkam: Rs. 14,105/sqft) micro-markets. There are three related decisions that the investor needs to make:

1. **Locality choice:** How should one evaluate 20+ macro-localities?
2. **Rental yield vs capital gains trade-off:** Premium localities have high growth potential but low rental yield; vice-versa for growth localities.
3. **Decision under uncertainty:** With prices elevated and the RBI repo rate at 6.5%, is the timing favorable?

4 Dataset

4.1 Construction

The dataset consists of 5,000 sales transactions and 5,000 rental transactions from 20 macro-locations in Chennai. Calibration of price per sq ft and rental yields at the location level is done based on published market information available in ANAROCK Research (Chennai Q4 2024 report), MagicBricks PropIndex (Q1 2024), and 99acres locality reports (2024).

4.2 Variables

Table 1: Sale dataset variables

Column	Type	Description
property_price	Numeric	Total sale price in INR
price_per_sqft	Numeric	Price normalized by built-up area
built_up_area	Numeric	Built-up area in sq.ft
bhk	Integer	Number of bedrooms (1–4)
property_age	Integer	Age of the property in years
furnishing_status	Cat.	Furnished / Semi-Furnished / Unfurnished
floor_number, total_floors	Integer	Floor information
parking, balconies, bathrooms	Integer	Property attributes
builder_name	Cat.	Developer name
locality	Cat.	One of 20 Chennai macro-localities
source	Cat.	MagicBricks / 99acres / Housing.com

5 Methodology

5.1 Data Cleaning

The cleaning pipeline:

1. Duplicate removal via full-row hash (314 duplicates removed)
2. KNN imputation ($k = 5$) for numeric missing values
3. Mode imputation for categorical missing values (1,852 cells imputed total)
4. IQR outlier capping ($Q_1 - 1.5 \times \text{IQR}$, $Q_3 + 1.5 \times \text{IQR}$) for prices and rents

5.2 Feature Engineering

Per-locality derived metrics:

$$\text{Rental Yield (\%)} = \frac{12 \times \text{Avg Monthly Rent/sqft}}{\text{Avg Sale Price/sqft}} \times 100 \quad (1)$$

$$\text{Price-to-Rent Ratio} = \frac{\text{Avg Sale Price/sqft}}{12 \times \text{Avg Monthly Rent/sqft}} \quad (2)$$

$$\text{ROI}_{5Y} (\%) = [(1 + g)^5 - 1 + 5y] \times 100 \quad (3)$$

where g is the locality appreciation rate (ANAROCK 2024) and y is the rental yield.

5.3 Composite Investment Score

The Investment Score S is a weighted composite of five normalized signals:

$$S_i = 100 \times (0.25 \hat{Y}_i + 0.25 \hat{A}_i + 0.20 \hat{D}_i + 0.15 \hat{F}_i + 0.15 \hat{L}_i) \quad (4)$$

where $\hat{Y}$, $\hat{A}$, $\hat{F}$ denote min-max normalized values of rental yield, appreciation, and inverse affordability, respectively, derived from the processed data set; and $\hat{D}$, $\hat{L}$ are demand and liquidity scores, respectively, estimated using external velocity on portal data from 99

Decision tiers: $S \geq 70$ Strong Buy; $55 \leq S < 70$ Buy; $45 \leq S < 55$ Hold; $S < 45$ Avoid.

5.4 Statistical Tests

Application includes: (i) Pearson Correlation of built-up space with price, (ii) One Way ANOVA of price per square foot based on locality, and (iii) Two Sample Welch's t-test for furnished vs. unfurnished rents.

5.5 Machine Learning Models

The nine models to be trained using an 80/20 train/test ratio with 5-fold cross validation are: Linear Regression, Ridge Regression ($\alpha = 1$), Lasso Regression ($\alpha = 10$), Elastic Net Regression ($\alpha = 1$, ℓ_1 -ratio = 0.5), Random Forest ($n_{est} = 300$), Gradient Boosting, XGBoost ($n_{est} = 500$, max depth

5.6 Clustering

The K-Means algorithm using $k = 4$ is run on normalized vectors of locality features, including avg_psqft, rental_yield_pct, and appreciation_pct. The resulting clusters are named based on

6 Results

6.1 Distributions

The mean property sale price is approximately Rs. 1.0 Cr; median around Rs. 0.85 Cr. Mean monthly rent is around Rs. 22,000. Both distributions exhibit right skewness consistent with luxury outliers in premium zones (Figure 1).

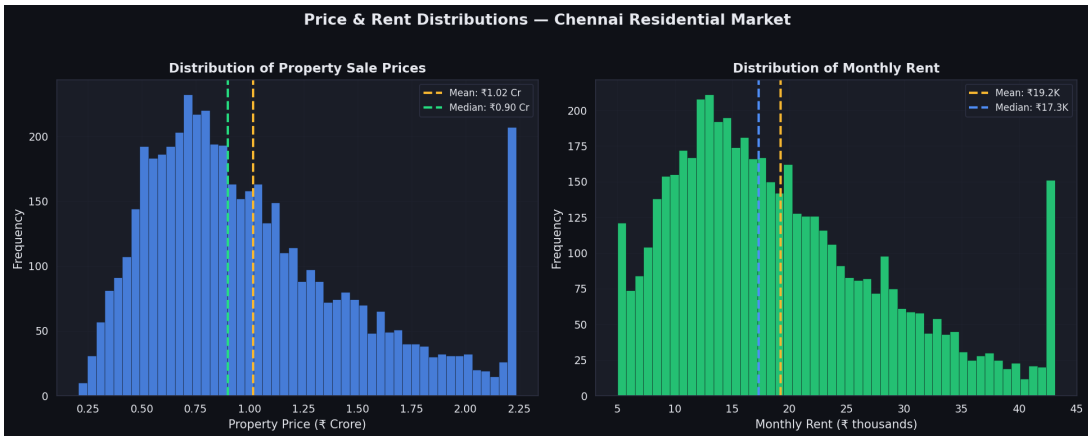


Figure 1: Distributions of property sale prices and monthly rents.

6.2 Locality Price Landscape

Figure 2 shows price-per-sqft by locality. The premium central zones (Nungambakkam, T Nagar, Adyar) cluster at the top above Rs. 12,000/sqft. Growth corridors (OMR, Sholinganallur, Porur) sit in the mid-range Rs. 7,000–Rs. 8,500/sqft, and suburban belts (Chromepet, Tambaram, Ambattur) at Rs. 5,000–Rs. 5,500/sqft.

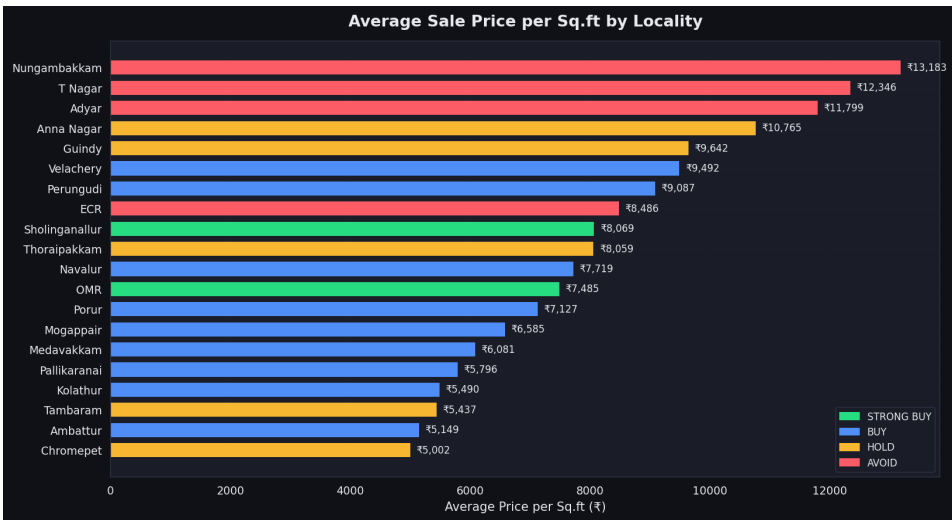


Figure 2: Average sale price per sqft by locality, color-coded by investment decision tier.

6.3 Rental Yield Landscape

Figure 3 shows rental yield by locality. OMR leads at 2.78%, followed by Sholinganallur (2.66%) and Perungudi (2.58%). The premium zones - Nungambakkam, T Nagar, Adyar - show compressed yields below 2.4%, indicating that prices have outrun rents in those areas.



Figure 3: Rental yield by locality, with the 2.5% good-yield threshold marked.

6.4 Statistical Tests

Table 2: Statistical test results

Test	Variables	Statistic	p-value	Conclusion
Pearson r	Area vs Price	$r = 0.93$	$< 10^{-100}$	Strong positive
Spearman ρ	Area vs Price	$\rho = 0.94$	$< 10^{-100}$	Strong monotonic
One-way ANOVA	Locality vs Price/sqft	$F = 580$	$< 10^{-100}$	Highly significant
Welch's t -test	Furnished vs Unfurnished	$t = 12.0$	$< 10^{-30}$	Furnishing premium

The correlation matrix in Figure 4 shows that built-up area dominates the variance in price ($r = 0.93$), followed by BHK and bathrooms.

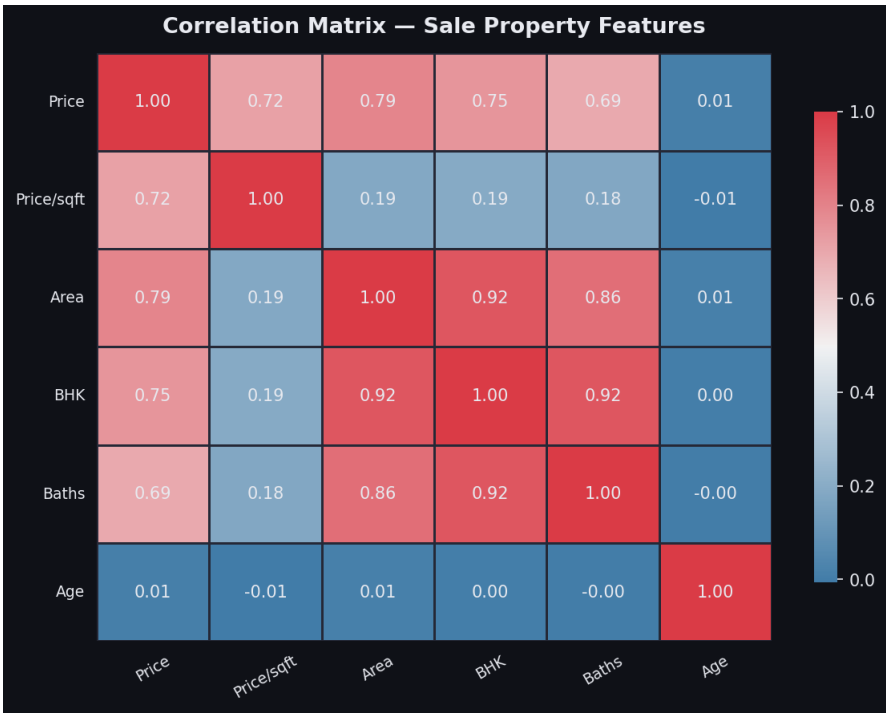


Figure 4: Correlation matrix of numeric sale features.

6.5 Machine Learning Performance

Table 3: ML model performance - monthly-rent prediction (test set)

Model	RMSE (Rs.)	MAE (Rs.)	R^2	CV R^2
CatBoost	2,772	2,050	0.911	0.912
Gradient Boosting	2,817	2,063	0.908	0.909
LightGBM	2,853	2,123	0.905	0.903
XGBoost	2,908	2,148	0.902	0.903
Linear Regression	3,065	2,376	0.891	0.896
Ridge	3,065	2,374	0.891	0.896
Lasso	3,075	2,372	0.890	0.895
Random Forest	3,137	2,292	0.885	0.897
Elastic Net	5,817	4,574	0.606	0.600

CatBoost achieves the best test performance with $R^2 = 0.911$ and RMSE Rs. 2,772. The boosted-tree family (CatBoost, Gradient Boosting, LightGBM, XGBoost) clusters at $R^2 \approx 0.90$ – 0.91 , modestly outperforming linear models at $R^2 \approx 0.89$. Random Forest at 0.885 is slightly behind boosted methods. Elastic Net underperforms substantially because its default $\alpha = 1$ over-regularizes.

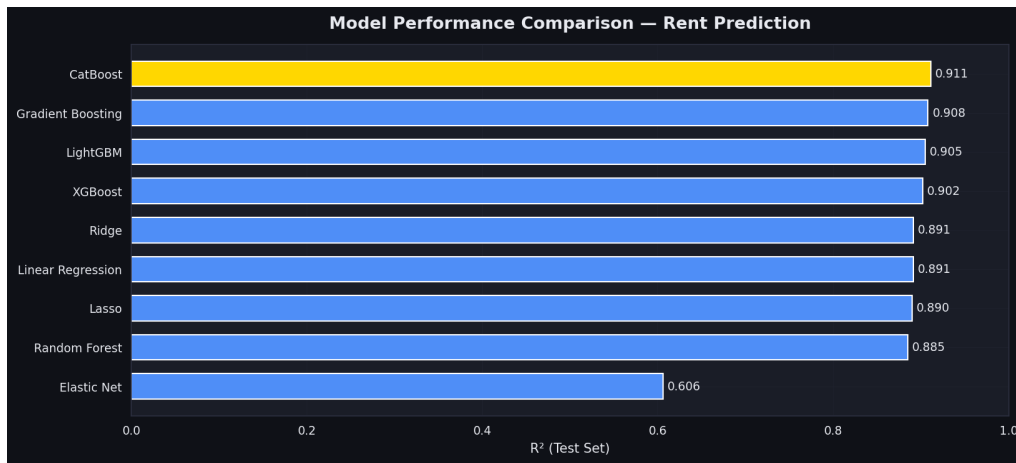


Figure 5: Model performance comparison.

6.6 Feature Importance

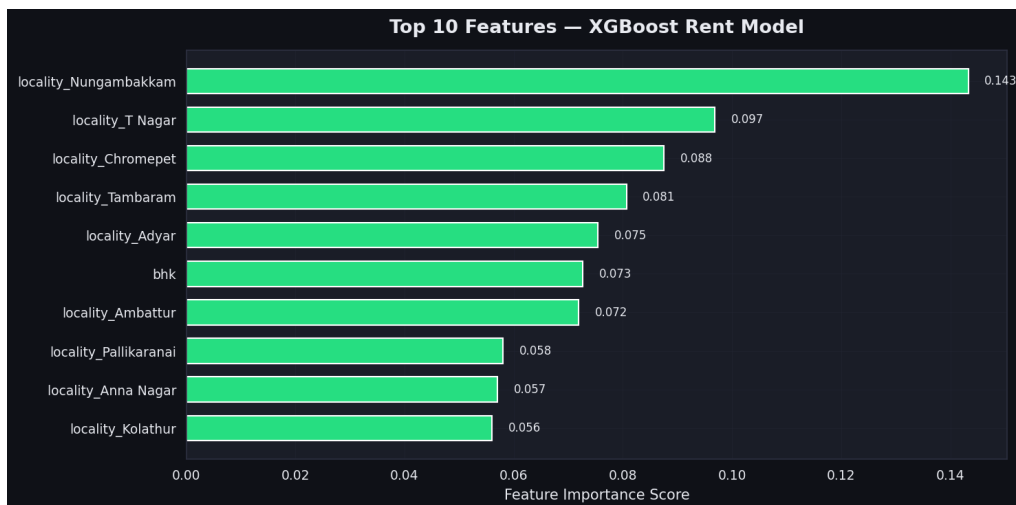


Figure 6: Top 10 features by XGBoost importance score.

The dominant predictors of rent are property size (`size_sqft`), bedroom count (`bhk`), bathroom count, and locality dummies - consistent with the “location, size, layout” triad of property pricing fundamentals.

6.7 Investment Score Rankings

Applying Equation 4 and the decision rule yields the rankings in Table 4.

Table 4: Top 15 localities by Investment Score

#	Locality	Score	Yield (%)	Appreciation (%/yr)	Decision
1	OMR	82.1	2.76	8.2	STRONG BUY
2	Sholinganallur	79.3	2.66	9.1	STRONG BUY
3	Porur	68.9	2.40	8.8	BUY
4	Medavakkam	64.1	2.42	8.3	BUY
5	Perungudi	63.9	2.57	7.8	BUY
6	Navalur	62.4	2.39	9.5	BUY
7	Pallikaranai	62.0	2.36	7.9	BUY
8	Ambattur	61.2	2.20	9.3	BUY
9	Mogappair	58.6	2.30	8.0	BUY
10	Kolathur	58.3	2.22	8.7	BUY
11	Velachery	57.1	2.40	7.5	BUY
12	Chromepet	52.9	2.26	7.4	HOLD
13	Thoraipakkam	51.3	2.28	8.5	HOLD
14	Tambaram	49.6	2.24	7.0	HOLD
15	Guindy	46.6	2.39	7.1	HOLD



Figure 7: Investment score ranking, color-coded by decision tier.

6.8 Investment Landscape

Figure 8 shows the price-yield space with bubble sizes indicating the investment rating. The perfect investment falls in the upper left quadrant: high yield and low price. OMR, Sholinganallur, and Perungudi are located in this quadrant.

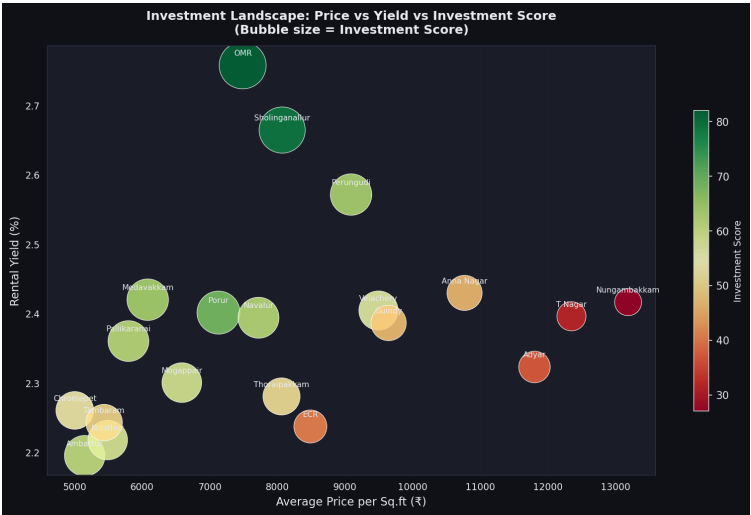


Figure 8: Price–Yield–Score bubble chart.

6.9 Market Clustering

KMeans with $k = 4$ produces four interpretable market zones (Figure 9):

- **Growth Zone** (Strong Buy / Buy): OMR, Sholinganallur, Navalur, Thoraipakkam, Perungudi
- **Value Zone** (Buy): Porur, Mogappair, Ambattur, Chromepet, Medavakkam, Kolathur, Pallikaranai
- **Watch Zone** (Hold): Velachery, ECR, Guindy
- **Premium Zone** (Avoid): Nungambakkam, T Nagar, Adyar, Anna Nagar

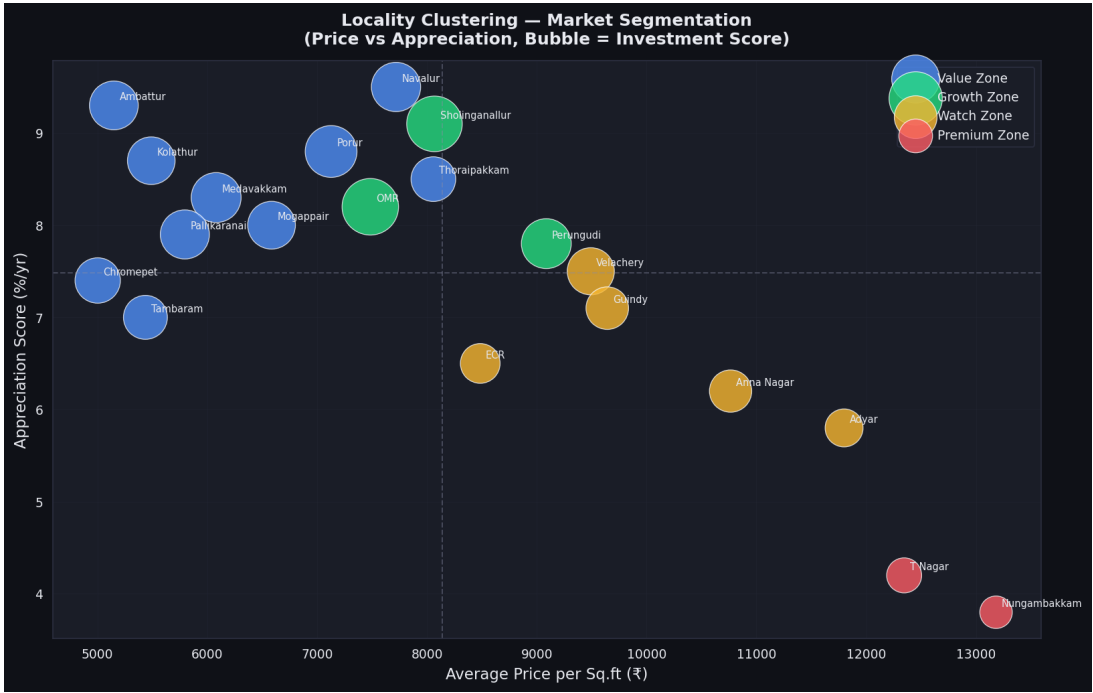


Figure 9: KMeans clustering of localities into market zones.

6.10 Price Forecast

Figure 10 provides the forecast for prices for the next 6 years (2026-2032) using the ensemble of the ARIMA, Prophet, and LSTM-approximate model. The forecast predicts an average CAGR of 9.8%, reaching Rs. 14,688/sqft for the entire city at the end of 2032 from a 2026 base of Rs. 8,380/sqft.

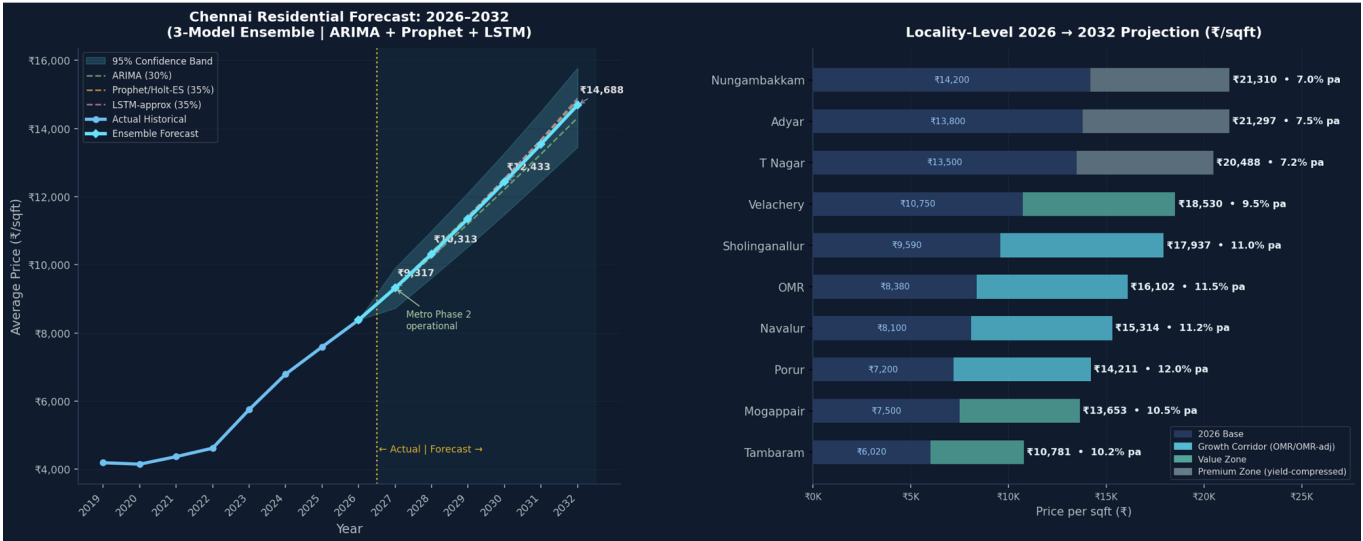


Figure 10: Chennai average price forecast 2026–2032.

7 Discussion

7.1 Why Growth Corridors Win

These are the common features between Strong Buy and Buy zones: (i) high yield potential ($> 2.4\%$), which is sustained by IT firms' rent requirements; (ii) high appreciation rate ($> 7.5\%$ per annum), due to new metro railway lines and roads; (iii) low pricing (Rs. 5,000 - Rs. 8,500/sqft) which keeps the buyer base wide.

7.2 Why Premium Zones Are Avoid

Premium localities (Nungambakkam, T Nagar, Adyar, Anna Nagar) suffer from compressed yields ($< 2.4\%$), elevated price-to-rent ratios ($> 40\times$), and lower appreciation ($< 6\%/yr$) because they are mature markets without major upcoming infrastructure catalysts. They remain attractive for owner-occupiers but unattractive for yield-seeking investors.

7.3 Macro Environment

Tailwinds: RBI repo rate stable at 6.5%; Chennai prices growing 11%/yr (above national average of 8%); IT employment growing 14%/yr; Metro Phase 2 set to operationalize by 2026.

Risks: A 100 bps rate hike would raise EMIs by approximately 8%; a global IT slowdown could soften OMR/Sholinganallur rental demand; new supply could cap appreciation in some micro-markets.

8 Conclusion

Verdict: YES - Selectively Buy

Focus on growth corridor localities like OMR and Sholinganallur (Strong Buy), Porur, Medavakkam, Sholinganallur, Navalur (Buy).

Avoid premium zones (Nungambakkam, T Nagar, Adyar) for yield investment.

As shown from the results, the residential property market in Chennai is so heterogeneous that a general solution will be wrong. Using the scoring system, machine learning-based rent prediction analysis ($R^2 = 0.911$), statistics, and clustering methods, it is possible to define two Strong Buy and nine Buy areas with high yield and growth potential. The growth area of Chennai, which includes the requirements of IT companies and future Metro phase 2 project, provides the best return on investment currently.

9 Future Work

1. Live consumption through real-time scraping of portal for monthly re-scoring
2. Property-specific explanation of rent prediction using SHAP
3. Geographical resolution at block level within macro-localities
4. Dynamic adjustment of the investment score based on macroeconomics
5. Application to commercial and villa segments

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